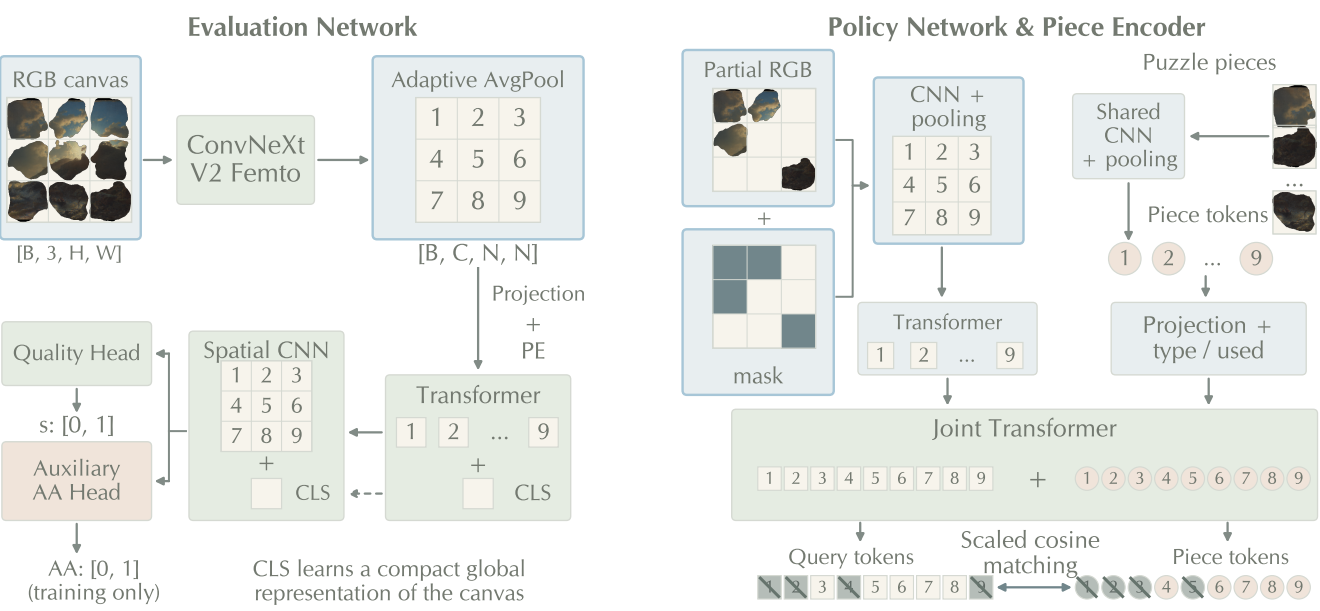

Evaluation Network
RGB canvas
[B, 3, H, W]
ConvNeXt V2 Femto
Adaptive AvgPool
1 2 3
4 5 6
7 8 9
[B, C, N, N]
Projection + PE
Transformer
1 2 ... 9
+
CLS
Spatial CNN
1 2 3
4 5 6
7 8 9
+
CLS
Quality Head
s: [0, 1]
Auxiliary AA Head
AA: [0, 1] (training only)
CLS learns a compact global representation of the canvas
Policy Network & Piece Encoder
Partial RGB
+
mask
CNN + pooling
1 2 3
4 5 6
7 8 9
Transformer
1 2 ... 9
Puzzle pieces
Shared CNN + pooling
Piece tokens
1 2 ... 9
Projection + type / used
Joint Transformer
1 2 3 4 5 6 7 8 9 + 1 2 3 4 5 6 7 8 9
Query tokens
Scaled cosine matching
Piece tokens